

ROTA & STAFF MANAGEMENT SOFTWARE

Complete Software Requirements & Functional Documentation

Document Version: 1.0

Purpose: Centralised staff, site, rota, shift, leave, document and timesheet management for security/staffing operations.

1. Project Overview

The Rota & Staff Management Software is a centralised web-based system designed to replace multiple Excel files and manual processes used for managing employees/guards, company sites, weekly rotas, shifts, leave, documents, timesheets and daily operations. The system will provide secure role-based access and a single source of operational information.

2. Objectives

- Create and maintain a single central database for employees, sites and operational records.
- Make weekly rota preparation faster and easier.
- Prevent scheduling conflicts caused by leave, unavailable staff or duplicate assignments.
- Track shifts, working hours, replacements and attendance-related information.
- Monitor employee documents and expiry dates.
- Provide management reports for working hours, staffing and operations.
- Reduce duplicate data entry, manual errors and dependence on separate Excel files.
- Support multiple authorised users with role-based permissions.
- Provide a scalable foundation for future mobile, payroll, client and notification features.

3. User Roles

4. Core Modules

4.1 Employee / Guard Management

- Create, edit, view and deactivate employee records.
- Store employee ID, name, photo, contact information, address and emergency contact.
- Store joining date, position, status and other configurable fields.
- Search by employee name or ID.
- Filter by status, site, role or other supported criteria.
- View previous shifts, sites and assignments.
- Maintain complete employee history in one place.
- View leave and document information from the employee profile.

4.2 Site Management

- Add, edit and manage company/client sites.
- Store site ID, site name, address and contact details.
- Define required number of guards and shift requirements.
- Configure day/night or custom shift requirements.

- Assign employees to specific sites.
- View current and historical assignments.

4.3 Weekly Rota Management

- Create weekly rotas from Monday to Sunday.
- Assign employees to sites and shifts.
- Support day and night shifts.
- Allow staff replacement and reassignment.
- Show unavailable employees and scheduling conflicts.
- Display unfilled shifts requiring action.
- View rota by employee, site or week.
- Publish/finalise a rota.
- Print and export weekly rota where required.

4.4 Shift Management

- Define shift start and finish times.
- Support day, night and custom shift patterns.
- Record scheduled and completed shifts.
- Handle shift changes and replacements.
- Maintain shift-change history.
- Calculate scheduled and actual working hours.

4.5 Leave & Vacation

- Record employee leave requests/records.
- Support configurable leave types such as annual, sick and emergency leave.
- View current and upcoming leave.
- Show leave on the rota/calendar.
- Prevent unavailable employees from being assigned to conflicting shifts.
- Maintain employee leave history.

4.6 Document Management

- Store employee document types and document numbers.
- Record issue and expiry dates.
- Upload documents or store secure external links.
- Support Google Drive-based document storage where required.
- Show valid, expiring and expired document status.
- Provide expiry reminders and management alerts.

4.7 Timesheet Management

- Record scheduled and actual working hours.
- Review completed shifts.
- Track overtime where applicable.
- Generate employee weekly/monthly hour summaries.
- Generate site-level hour summaries.

- Provide data for salary and client invoice calculations.

4.8 Daily Operations

- Provide a Today Operations view.
- Show today's sites, shifts and assigned employees.
- Record attendance/follow-up status where required.
- Identify missing/unconfirmed staffing.
- Record operational notes and issues.
- Highlight urgent replacement or staffing requirements.

4.9 User Access & Permissions

- Secure login and logout.
- Role-based access control.
- Module-level permissions.
- Optional create/read/update/delete permissions per role.
- Prevent unauthorised access to employee and operational data.

5. Dashboard Requirements

- Total active employees/guards.
- Total active sites.
- Today's shifts.
- Present/absent/unconfirmed staffing indicators.
- Employees currently on leave.
- Unfilled shifts.
- Documents expiring soon.
- Expired documents.
- Quick links to create rota, employee, site or leave record.
- Operational alerts and notifications.

6. Rota Rules & Business Logic

- An employee should not be assigned to a shift during approved leave.
- The system should warn or prevent overlapping shifts for the same employee.
- A site should show whether its required staffing level is fulfilled.
- Replacing an employee should preserve the original assignment history.
- Shift changes should be recorded with user, date/time and reason where enabled.
- Night shifts crossing midnight must be handled correctly for date and working-hour calculations.
- Inactive employees should not normally appear in new rota assignment lists.
- Rota changes should be auditable.

7. Notifications & Alerts

- Document expiry reminders.
- Upcoming leave reminders.

- Unfilled shift alerts.
- Staff replacement alerts.
- Rota publication/update notifications.
- Optional email or messaging integration in future versions.

8. Reports

- Weekly rota report.
- Employee working-hours report.
- Site working-hours report.
- Overtime report.
- Leave report.
- Absence/attendance report.
- Document expiry report.
- Staffing gap report.
- Payroll preparation summary.
- Client invoice preparation summary.

9. Audit Log

- Record important create/update/delete actions.
- Store the responsible user.
- Store date and time.
- Record relevant before/after changes where appropriate.
- Track employee assignment and replacement changes.

10. Centralised Database

The system should maintain related data in a structured relational database. Suggested core entities include:

- Users
- Roles & Permissions
- Employees
- Employee Documents
- Sites
- Shift Types
- Rota Weeks
- Rota Assignments
- Shifts
- Leave Records
- Timesheets
- Attendance/Operational Records
- Notifications
- Audit Logs

11. Suggested Technology Stack

- Frontend: React.js
- UI: Tailwind CSS
- Backend: Django + Django REST Framework
- Database: PostgreSQL
- Authentication: JWT or secure session authentication
- Permissions: Role-Based Access Control (RBAC)
- Document Storage: Google Drive integration and/or secure cloud storage
- PDF/Print: Server-side PDF generation and print-friendly views
- Deployment: Railway, Render, VPS or equivalent production hosting

12. Security Requirements

- Password-protected user accounts.
- Secure authentication and authorisation.
- Role-based permissions.
- HTTPS in production.
- Server-side validation of all important data.
- Protection of employee personal information.
- Secure document access.
- Audit logging for sensitive operational changes.
- Regular database backups in production.

13. UI / UX Requirements

- Responsive design for desktop, tablet and mobile.
- Simple navigation with clear modules.
- Fast global search.
- Clear status badges for leave, documents and staffing.
- Calendar and weekly rota views.
- Confirmation dialogs for destructive actions.
- Print-friendly rota layout.
- Accessible forms with validation and helpful error messages.

14. Future Expansion

- Employee mobile application.
- Guard self-service portal.
- Client portal.
- GPS/geolocation-based attendance where legally and operationally appropriate.
- Biometric attendance integration.
- SMS/WhatsApp notifications.
- Payroll automation.
- Automated client invoicing.

- Advanced analytics and dashboards.
- AI-assisted rota optimisation.
- Multi-company/multi-branch support.

15. Key Benefits

- Saves administrative time.
- Reduces manual work and human error.
- Makes weekly rota preparation easier.
- Provides quick access to employee information.
- Improves staff and site management.
- Helps prevent document expiry issues.
- Improves management control.
- Organises daily operations.
- Supports multiple authorised users.
- Provides a scalable platform for future features.

16. Recommended Development Phases

Phase 1 – Foundation: Authentication, users/roles, database structure, dashboard and core settings.

Phase 2 – Employee & Site: Employee/guard management, site management and assignment history.

Phase 3 – Rota & Shifts: Weekly rota, shift patterns, assignments, replacements and conflict checking.

Phase 4 – Leave & Documents: Leave management, document records, expiry tracking and alerts.

Phase 5 – Timesheets & Operations: Completed shifts, working hours, daily operations and summaries.

Phase 6 – Reports & Export: Reports, PDF/print views, Excel export and management summaries.

Phase 7 – QA & Deployment: Testing, security review, performance checks, deployment, backups and documentation.

17. Acceptance Criteria – MVP

- Admin can securely log in and manage users.
- Admin/Manager can create and manage employees and sites.
- Manager can create a weekly rota and assign employees to shifts/sites.
- The system warns against leave conflicts and overlapping assignments.
- Managers can replace assigned staff while preserving history.
- Leave records are visible and affect rota availability.
- Employee documents can be recorded with expiry dates.
- Timesheets can calculate working-hour summaries.
- Daily operations show current staffing requirements.
- Authorised users can generate/print a weekly rota.
- Basic reports are available for employees, sites, hours, leave and documents.
- Core actions are protected by role-based permissions.

18. Conclusion

The proposed Rota & Staff Management Software will provide a single, organised and secure platform for managing employees/guards, sites, rotas, shifts, leave, documents, timesheets and daily operations. The architecture should be modular so that payroll, client invoicing, mobile applications, notifications and advanced automation can be added without rebuilding the core system.

Role	Access	Main Functions	Restrictions
Admin	Full	All modules, users, settings, reports, configuration	None
Manager	High	Employees, sites, rota, shifts, leave, timesheets, reports	Cannot manage selected system/security settings unless permitted
Receptionist	Limited	View employees/sites/rota; permitted operational entry	View-only or restricted editing based on permissions